

NETWORKS AND COMPLEXITY

Solution 23-4

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 23.4: Customers [Lvl. 3]

In a shop customers arrive at an average rate of a customers per minute. Customers in the shop leave at a rate of b customers per customer per minute. Use detailed balance to compute x_n the probability that there are n customers in the shop, in the stationary state.

Solution

We define x_i as the probability that there are i customers in the shop starting from $i = 0$. In principle, there is now an infinity of microstates, which makes using Kirchhoff's approach complicated. However, since the number of customers only changes by one in each event, the microstates form a chain, (0)-(1)-..., so that we can use detailed balance.

Let's start by considering the link between 0 and 1 customers in the shop. If there is just one customer in the shop, they will leave at rate b . If there are no customers in the shop, one will arrive at rate a . Hence, the detailed balance condition between states 0 and 1 is

$$ax_0 = bx_1 \quad (1)$$

and hence

$$x_1 = \frac{a}{b}x_0 \quad (2)$$

Now consider the detailed balance condition between the states 1 and 2. If there is one customer in the shop, a second will arrive at rate a , and if there are two customers each of them will leave at rate b , such that the rate of one of them leaving is actually $2b$. So, the detailed balance condition is

$$ax_1 = 2bx_2 \quad (3)$$

and hence

$$x_2 = \frac{a}{2b}x_1 = \frac{a^2}{2b^2}x_0 \quad (4)$$

For the link between microstate 2 and 3 we have the detailed balance condition

$$ax_2 = 3bx_3 \quad (5)$$

and hence

$$x_3 = \frac{a}{3b}x_2 = \frac{a^3}{6b^3}x_0 \quad (6)$$

Let's generalize to state x_n . The detailed balance condition between n and $n - 1$ is

$$ax_{n-1} = nbx_n \quad (7)$$

and hence

$$x_n = \frac{a}{nb} x_{n-1} \quad (8)$$

$$= \frac{a}{nb} \frac{a}{(n-1)b} x_{n-2} \quad (9)$$

$$= \left(\prod_{m=1}^n \frac{a}{mb} \right) x_0 \quad (10)$$

$$= \frac{a^n}{b^n n!} x_0 \quad (11)$$

Let's define

$$r = \frac{a}{b} \quad (12)$$

This allows us to write

$$x_n = \frac{r^n}{n!} x_0 \quad (13)$$

To find the value of x_0 we need to check the normalization. We check

$$1 = \sum_{n=0}^{\infty} \frac{r^n}{n!} x_0 = x_0 \sum_{n=0}^{\infty} \frac{r^n}{n!} = x_0 e^r \quad (14)$$

OK, that's convenient, so

$$x_0 = e^{-r} \quad (15)$$

and

$$x_n = \frac{e^{-r} r^n}{n!} \quad (16)$$

Wait a minute, does this remind you of something? (Say, $p_k = e^{-z} z^k / k!$).

Alright, yes, this a Poisson distribution. So, the number of customers in the shop will be Poisson distributed with a mean $r = a/b$.